

MATTHEW T. COPE

Warner Robins, GA | 434-238-3868 | mtcope95@gmail.com

C++ SOFTWARE DEVELOPMENT | SYSTEMS & TEST | TECHNICAL LEADERSHIP

B.S. Computer Science Candidate - Expected March 2027 | Active TS/SCI Eligibility

PROFESSIONAL SUMMARY

Computer science student with a 4.0 GPA and 13 years of experience in technical operations, evaluation, instruction, and aerospace systems. Hands-on C++ development experience includes object-oriented design, modular architecture, interactive 2D and 3D graphics, collision detection, debugging, and Git version control. Also experienced with Java, Python, requirements-based testing, systems troubleshooting, and translating technical standards into reliable solutions for complex operational environments.

TECHNICAL QUALIFICATIONS

Programming: C++, Java, Python, and MATLAB; object-oriented design, modular architecture, data structures, and algorithms

Software Engineering & Test: Requirements analysis, design, implementation, unit testing, edge-case testing, debugging, verification, refactoring, code coverage, and software quality assurance

Development Tools: Git, GitHub, Visual Studio, VS Code, Eclipse, Jupyter, OpenGL, GLFW, GLAD, and GLM

Systems & Engineering: Root-cause analysis, technical-data interpretation, hardware troubleshooting, corrective action, standards compliance, 2D/3D rendering, transformations, lighting, textures, and camera systems

Professional Strengths: Technical documentation, stakeholder coordination, instruction, risk identification, performance feedback, cross-functional leadership, and senior-leader advisement

EDUCATION

Southern New Hampshire University | B.S. Computer Science, Software Engineering Concentration

GPA: 4.0 | Expected Graduation: March 2027 | Relevant Coursework: Software Engineering, Software Testing, Data Structures & Algorithms, Computer Graphics & Visualization, Applied Linear Algebra, Statistics

Community College of the Air Force | A.A.S. Aircraft Maintenance Technology, 2025

SELECTED TECHNICAL PROJECTS

3D Interactive Graphics Application | C++ / OpenGL

- Developed an interactive 3D application using reusable meshes, object transformations, textures, directional and point lighting, and keyboard and mouse camera controls.
- Diagnosed and corrected rendering, texture, lighting, and object-placement defects through iterative debugging and testing, improving visual accuracy and application stability.

Breakout-Style 2D Application | C++ / Object-Oriented Design

- Refactored a monolithic graphics program into multiple classes and source files, improving modularity, maintainability, separation of responsibilities, and code reuse.
- Implemented and debugged multi-object collision detection, randomized behavior, multi-ball interactions, and corrective movement logic to prevent stalled execution paths.

Software Testing & Quality Assurance | Java / JUnit

- Translated requirements into positive, negative, boundary, validation, immutability, and date-rule tests for contact, task, and appointment services.
- Achieved 87.7% code coverage, exceeding an 80% target, while documenting defects and refactoring code to strengthen reliability and maintainability.

PROFESSIONAL EXPERIENCE

Technical Instructor / Evaluator (C-146A Formal Training Unit) | U.S. Air Force Reserve, 5th Special Operations Squadron | Feb 2026-Present
Duke Field, FL | 5 hrs/week

- Deliver standardized technical instruction and verify performance against qualification, safety, and mission-readiness requirements; identify gaps, document results, and provide actionable corrective feedback.
- Collaborate with instructors, evaluators, operations personnel, and leaders to resolve technical, qualification, and scheduling issues while maintaining accurate program information.

Operations Manager / Technical Evaluator | U.S. Air Force, 492d Special Operations Wing | Jan 2025-Jul 2025
Duke Field, FL | 45 hrs/week

- Led daily operations for 40 personnel, analyzing qualifications, requirements, constraints, and program risks to support timely decisions in a complex special-operations environment.
- Evaluated personnel and programs against technical standards, documented findings, analyzed deficiencies, and advised stakeholders on corrective actions to improve compliance and readiness.

Program Manager / Technical Instructor (Standardization & Evaluation) | U.S. Air Force, 492d Special Operations Wing | Sep 2023-Jan 2025
Duke Field, FL | 45 hrs/week

- Led a technical standardization and evaluation program for 40 personnel; interpreted requirements, reviewed records and programs, identified trends, and developed data-informed improvement recommendations.
- Directed Unit Effectiveness Inspection preparation by coordinating requirements, records, and stakeholders, contributing to zero discrepancies from wing and Air Force Special Operations Command evaluators.

Deployment Readiness Manager / Technical Instructor | U.S. Air Force, 524th Special Operations Squadron | May 2021-Sep 2023
Duke Field, FL | 45 hrs/week

- Managed readiness requirements for 120 personnel, tracked technical and documentation status, identified deficiencies, and coordinated corrective actions across multiple organizations.

Aircraft Systems Technician (Aircraft Fuel Systems) | U.S. Air Force, 366th Maintenance Group | Dec 2013-Dec 2019
Mountain Home AFB, ID | 45 hrs/week

- Diagnosed and repaired aircraft-system discrepancies using technical orders, schematics, inspection procedures, and systematic troubleshooting; documented maintenance, conducted hands-on hardware testing, and verified system performance.

CLEARANCE & RECOGNITION

- Active TS/SCI eligibility documented in Air Force personnel records; able to obtain and maintain a DoD Secret clearance.
- Master Sergeant (E-7), U.S. Air Force Reserve; 13 years of service and five contingency deployments.
- Wing Standardization/Evaluation NCO of the Year, 2024.